Sahaj Bhadja

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Summary

Software Engineer pursuing an M.S. in Computer Science at USC with 2+ years building **business application systems**, **SQL-backed data workflows**, and reliable **production support/troubleshooting** pipelines using strong SDLC, testing, and cross-functional collaboration practices.

Education

University of Southern California **Los Angeles, CA**
Master of Science in Computer Science *August 2024 – May 2026*
GPA: 3.75/4 Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University **Gandhinagar, India**
B.Tech. in Computer Science and Engineering *November 2020 – May 2024*
GPA: 3.88/4 Coursework: OOP, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Built **Python** workflow services and data interfaces for **500+ users**, reducing latency from **1.2s to 350ms** through query optimization while improving application reliability and production support readiness for critical business workflows.
- Implemented fault-tolerant orchestration for **5K+ monthly jobs** with validation safeguards and rollback checks, improving **issue triage speed** and reducing incident impact across stage, test, and production-like environments.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Developed **React** and **Flask** features handling **10K+ daily requests**, translating business requirements into maintainable application modules and secure APIs while improving documentation clarity and QA collaboration.
- Improved release velocity with **CI/CD** controls and concurrent processing, reducing turnaround time by **60%** while preserving quality through automated checks, regression-focused testing, and structured defect-resolution workflows.

USC AI Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **Python** experimentation pipelines over **100K+ trajectory logs**, analyzing system behavior and anomaly patterns that improved root-cause investigations and strengthened data-driven debugging for complex applications.
- Produced technical summaries and diagnostics that translated findings into implementation plans, improving **cross-team communication** and enabling faster incident resolution across research, engineering, and operations stakeholders.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **API** and **ETL** microservice workflows with OOP, AWS Glue, and S3, improving response times by **30%** while supporting scalable integrations and dependable data exchange for enterprise applications.
- Maintained **20+** unit and integration suites using Jenkins, PyTest, and Mockito across **Linux/Windows** environments, boosting pipeline efficiency by **68%** and reducing technical debt by **20%** through automation.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **REST API** and **WebSocket** modules supporting eight workflows and **2M+ synchronized records**, improving application stability and user-facing responsiveness in high-throughput operations environments.
- Automated deployments and testing with **Docker** and **Jenkins**, reducing deployment time by **50%** and improving release consistency through scripted validations, rollback planning, and incident-prevention checks.

Skills

Languages: Python, JavaScript, TypeScript, Java, C++, SQL, Bash
Application Development: Flask, FastAPI, REST APIs, Business Workflow Automation, Report-Oriented Data Processing
Data & Systems: SQL Development, Relational Databases, ETL Pipelines, Unix/Linux Environments, Shell Scripting
Cloud & Tooling: AWS (S3, Glue, ECS), Docker, Kubernetes, Jenkins, Git, CI/CD
Engineering Practices: SDLC, Unit/Integration Testing, Incident Troubleshooting, Technical Documentation, Agile/Scrum

Projects

HMIS | *MERN, ABDM APIs, Twilio, Sentry, TypeScript*

- Built a healthcare operations platform integrating **ABDM APIs** and **Sentry** monitoring, reducing administrative workload by **30%** through structured workflows, improved issue visibility, and reliable production support processes.

Chat2SQL | *Java, TypeScript, SQL, Docker, Maven*

- Built an AI-assisted SQL workspace supporting **10+ databases**, improving query authoring and schema exploration through maintainable data-access workflows, repeatable validations, and business-application troubleshooting capabilities.

Linguistix | *Python, AWS, Docker, CI/CD*

- Developed and deployed a containerized **RESTful API** on AWS CodePipeline and ECS/Fargate with automated **CI/CD** and validation checks, achieving **87% accuracy** and production-grade reliability under sustained workloads.